TimeLine
of
SUFFERING
2008-2009
2002
2023
NO MORE
لا مزيد
IMPACT OF WAR:
2000-2023

ISRAELI–PALESTINIAN CONFLICT FATALITIES

2000 – 2023

Exploratory Data Analysis

Analyst: Monsurat Olayinka Onashokun

DataSmart Point · Python for Data Analysis – Final Project

11,124

Total Fatalities

11,083

Unique Individuals

23 Years

Data Coverage

16 Columns

Features

AGENDA

01 Research Question & Dataset Overview

02 Data Source & Quality

03 Data Cleaning & Feature Engineering

04 Univariate Analysis

05 Bivariate & Multivariate Analysis

06 Temporal Analysis — The July Finding

07 Key Findings & Conclusion

08 Limitations & Reflection

01 · RESEARCH QUESTION & DATASET OVERVIEW

"Is the intensity of the Israeli–Palestinian conflict increasing over time,

and which years, regions and causes show the highest fatality rates?"

Dataset	Format	Coverage	License
B'Tselem Fatality Database Israeli human rights organisation	CSV file · 11,124 rows 16 columns per record	January 2000 – December 2023	Open data · publicly available for research

KEY COLUMN TYPES

age	Metric / Continuous	date_of_event	Temporal
citizenship	Nominal Categorical	event_location_region	Nominal Categorical
gender	INominal Binary	type_of_injury	Nominal Categorical

02 · DATA CLEANING & FEATURE ENGINEERING

FIX A

Column Rename

Renamed took_part_in_the_hostilities → hostility_status for code-safe access without KeyError crashes.

FIX C

Date Parsing

Parsed date_of_event to datetime. Extracted year column for time-series analysis across all 23 years.

FIX B

Age Data Type

Converted age from object → numeric. Filled 129 NaN values using median imputation (avoids skew from outliers).

FIX D

Categorical Nulls

Filled missing values in gender, type_of_injury, ammunition → 'Unknown' to prevent chart gaps.

FEATURE ENGINEERING

life_stage	Bins age into Child / Teenager / Young Adult / Adult / Middle-Aged / Senior
is_minor / is_senior	Binary flags (1/0) for fast vulnerability group filtering
is_age_outlier	IQR-based outlier detection — identifies extreme ages statistically
injury_category	Custom function categorizes raw text into 7 clean injury groups
ammo_category	Custom function groups 50+ ammunition types into 6 categories

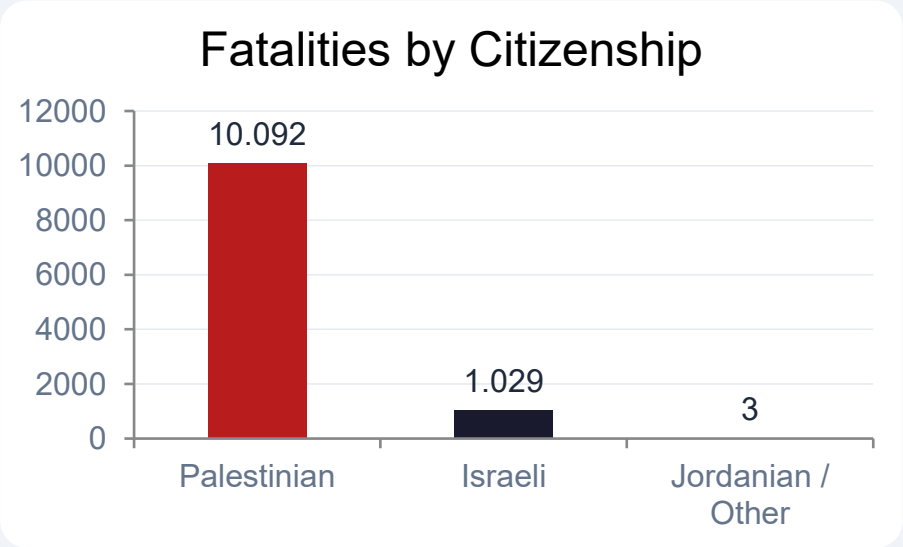
03 · UNIVARIATE ANALYSIS — Who Are the Victims?

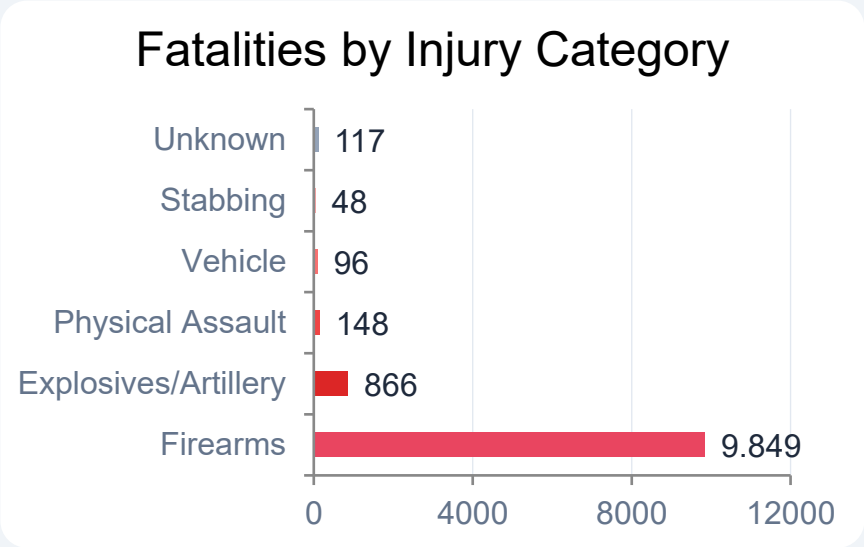
10,092
Palestinian Fatalities
(90.7%)

1,029
Israeli Fatalities
(9.2%)

87.1%
Male Victims

20.1%
Victims Under 18





Firearms account for 88.5% of all deaths. Palestinian fatalities outnumber Israeli by a ratio of nearly 10:1.

04 · BIVARIATE ANALYSIS — Conflict Intensity Over Time

2014 — Peak Year

2,332 total fatalities during Operation Protective Edge

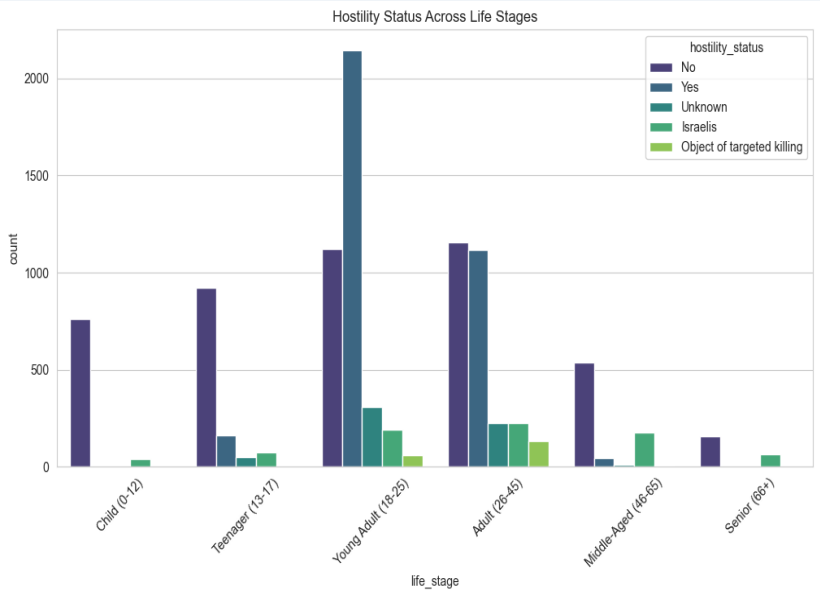
2009 — Most Asymmetric

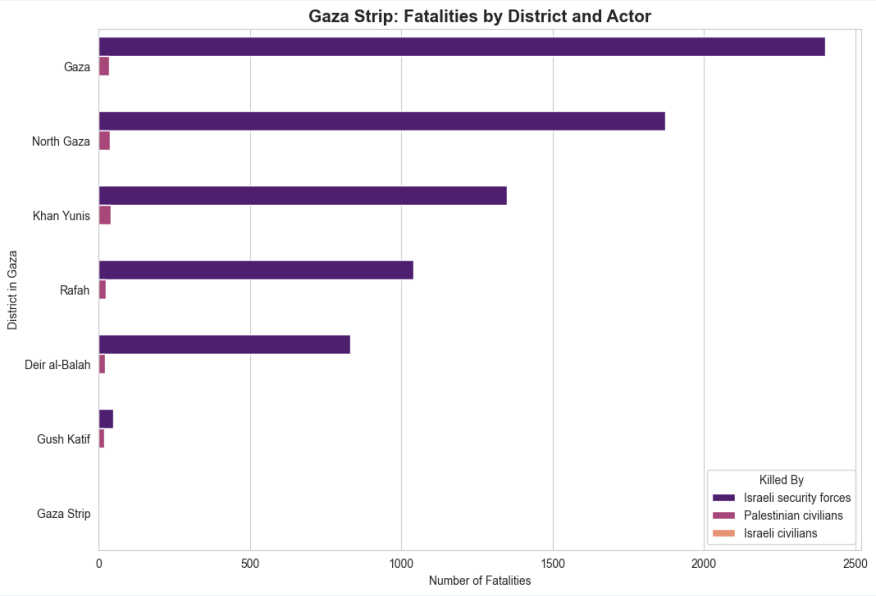
115:1 ratio (1,036 Palestinian vs only 9 Israeli)

Pattern

Spikes — not steady growth.
Violence is cyclical, not linear.

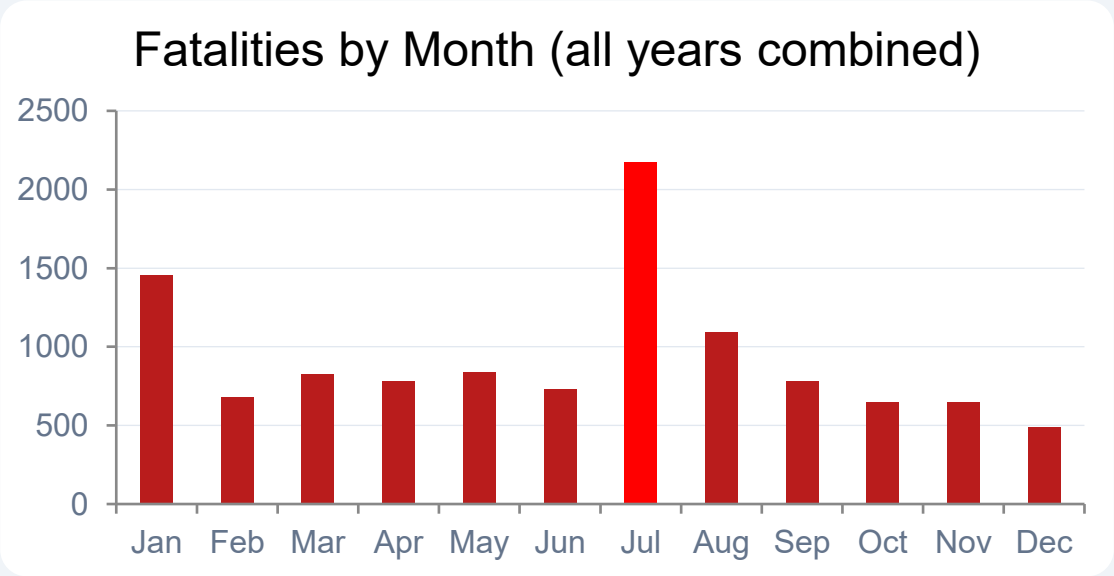
05 · BIVARIATE ANALYSIS — Region, Life Stage & Tactics





- Gaza accounts for 69.5% of all fatalities — driven by dense urban population and large-scale military operations.
- Young Adults (18–25) are the most affected life stage (3,832), reflecting frontline exposure.
- 2,239 victims (20.1%) were under 18 — confirming significant civilian / child impact across all regions.

06 · TEMPORAL ANALYSIS — The July Finding



⚠️ The July Trap

July shows 2,173 total fatalities — the highest of any month.

Operation Protective Edge
Gaza · July 2014

Without July 2014, monthly distributions are nearly even.

📊 Key Data Analysis Lesson

Aggregate totals can hide the true cause behind a pattern. July appears deadly across 23 years — but it is dominated by ONE month in ONE year. This is why drilling down before drawing conclusions is essential in any data analysis.

07 · KEY FINDINGS

1 Cyclical, Not Linear

Violence does not grow steadily. It surges in specific military operations: 2002 (Intifada), 2009 (Cast Lead), 2014 (Protective Edge).

2 Extreme Asymmetry

Palestinians: 10,092 (90.7%) · Israelis: 1,029 (9.2%). Ratio of 9.8:1 — consistent across all years and regions.

3 Firearms Dominate

88.5% of all deaths caused by gunfire. Explosives spike in Gaza during major offensives (2008, 2009, 2014).

4 The July 2014 Trap

July appears deadliest but 77% of its total comes from one single month in 2014. Aggregate data misleads without drill-down.

5 Children & Women

2,239 victims (20.1%) were under 18. 1,423 (12.8%) were female. Civilians are a significant share of fatalities.

6 Gaza is the Epicentre

69.5% of all fatalities in Gaza Strip. 10:1 child fatality ratio (Gaza vs West Bank). Dense urban conflict zone.

08 · LIMITATIONS & REFLECTION

① **Source Bias**

Data from B'Tselem (Israeli NGO focused on Palestinian rights). Israeli military fatalities outside B'Tselem's scope may be underrepresented.

② **Missing Data — hostility_status**

~60% of the 'combatant vs civilian' column is unknown. This makes it impossible to draw reliable conclusions about combatant ratios.

③ **Correlation ≠ Causation**

The data shows WHEN and WHERE deaths occurred — but cannot explain WHY. Political decisions and international events are outside this dataset's scope.

④ **Incomplete 2023 Data**

The October 7, 2023 attack and Gaza response are only partially captured. A full 2023 analysis would require updated data.

FINAL ANSWER

"Is the intensity of the conflict increasing over time?"

Not in a linear sense — but YES in a cyclical sense.

Each decade has produced at least one extreme escalation. 2014 produced more fatalities in a single month (July: 1,673) than entire years in the early 2000s. The conflict has not de-escalated — it has become concentrated into shorter, more intense bursts of extreme violence separated by periods of lower but continuous fatalities.

11,124	9.8:1	88.5%	20.1%
Total Fatalities	Palestinian:Israeli Ratio	Deaths by Firearms	Victims Under 18

Thank you

Monsurat Olayinka Onashokun